

Table S2. Context-specific predictive performance of chemical and transcriptome-complemented models across the twelve Tox21 endpoints. AUPRC is reported to three decimal places for Chem. and Chem. + Bio. on matching held-out compounds within each endpoint, cell line, and exposure time. The 6-h rows represent primary full cohorts; the 24-h rows represent paired exposure-time cohorts. The frozen chemical logit is calibrated on each training partition; this intercept is held fixed while transcriptomic β coefficients are fitted using unweighted BCE plus a ridge penalty selected by inner cross-validation. Unadjusted perturbation expression is standardized using training-fold means and standard deviations. Bold Chem. + Bio. values and asterisks denote nominally significant positive improvements over the corresponding Chem. value (*P < 0.05, two-sided Nadeau-Bengio corrected resampled test, without multiple-testing adjustment). Endpoint labels are bold when at least one displayed context meets this criterion. † denotes fewer than 20 active compounds. Paired-cohort 6-h results are retained in the full numeric source but are not displayed.

Endpoint	Cell	Time (h)	n	Active	Chem.	Chem. + Bio.
NR-AR	HA1E	6	968	72	0.461	0.461
NR-AR	HepG2	6	668	46	0.537	0.537
NR-AR	HT29	6	960	72	0.457	0.457
NR-AR	MCF7	6	1259	82	0.459	0.459
NR-AR	HA1E	24	830	59	0.432	0.432
NR-AR	HT29	24	585	44	0.385	0.385
NR-AR	MCF7	24	1243	78	0.460	0.460
NR-AR-LBD	HA1E	6	869	49	0.554	0.554
NR-AR-LBD	HepG2	6	598	37	0.554	0.554
NR-AR-LBD	HT29	6	862	47	0.561	0.561
NR-AR-LBD	MCF7	6	1145	53	0.581	0.585
NR-AR-LBD	HA1E	24	747	39	0.503	0.503
NR-AR-LBD	HT29	24	523	19	0.510	0.510†
NR-AR-LBD	MCF7	24	1129	51	0.583	0.586
NR-AhR	HA1E	6	849	102	0.526	0.526
NR-AhR	HepG2	6	585	65	0.477	0.482
NR-AhR	HT29	6	841	95	0.479	0.479
NR-AhR	MCF7	6	1111	112	0.473	0.472
NR-AhR	HA1E	24	727	91	0.527	0.530
NR-AhR	HT29	24	516	63	0.521	0.521
NR-AhR	MCF7	24	1099	111	0.468	0.468
NR-Aromatase	HA1E	6	759	63	0.384	0.384
NR-Aromatase	HepG2	6	512	47	0.346	0.356
NR-Aromatase	HT29	6	758	63	0.372	0.378
NR-Aromatase	MCF7	6	1021	76	0.409	0.431
NR-Aromatase	HA1E	24	655	57	0.397	0.482
NR-Aromatase	HT29	24	475	45	0.371	0.371
NR-Aromatase	MCF7	24	1012	76	0.411	0.538*
NR-ER	HA1E	6	805	132	0.430	0.430
NR-ER	HepG2	6	540	92	0.477	0.477
NR-ER	HT29	6	799	128	0.419	0.419
NR-ER	MCF7	6	1067	155	0.402	0.402
NR-ER	HA1E	24	689	112	0.441	0.441
NR-ER	HT29	24	487	76	0.409	0.409
NR-ER	MCF7	24	1054	154	0.407	0.407
NR-ER-LBD	HA1E	6	907	55	0.385	0.384
NR-ER-LBD	HepG2	6	620	39	0.404	0.404
NR-ER-LBD	HT29	6	902	53	0.397	0.398
NR-ER-LBD	MCF7	6	1194	61	0.379	0.418
NR-ER-LBD	HA1E	24	772	40	0.411	0.400
NR-ER-LBD	HT29	24	547	25	0.375	0.369
NR-ER-LBD	MCF7	24	1178	59	0.397	0.429
NR-PPAR-gamma	HA1E	6	811	34	0.235	0.235
NR-PPAR-gamma	HepG2	6	552	19	0.300	0.299†
NR-PPAR-gamma	HT29	6	803	29	0.193	0.192
NR-PPAR-gamma	MCF7	6	1073	34	0.222	0.222
NR-PPAR-gamma	HA1E	24	696	29	0.241	0.241
NR-PPAR-gamma	HT29	24	483	17	0.278	0.277†

Table S2 continued

Endpoint	Cell	Time (h)	n	Active	Chem.	Chem. + Bio.
NR-PPAR-gamma	MCF7	24	1060	33	0.212	0.212
SR-ARE	HA1E	6	690	136	0.474	0.479
SR-ARE	HepG2	6	451	98	0.452	0.553*
SR-ARE	HT29	6	682	129	0.451	0.452
SR-ARE	MCF7	6	923	160	0.452	0.459
SR-ARE	HA1E	24	584	118	0.469	0.482
SR-ARE	HT29	24	416	80	0.498	0.492
SR-ARE	MCF7	24	913	159	0.449	0.463
SR-ATAD5	HA1E	6	911	50	0.295	0.293
SR-ATAD5	HepG2	6	619	33	0.372	0.368
SR-ATAD5	HT29	6	903	50	0.307	0.316
SR-ATAD5	MCF7	6	1196	57	0.254	0.260
SR-ATAD5	HA1E	24	778	43	0.311	0.337
SR-ATAD5	HT29	24	544	30	0.308	0.308
SR-ATAD5	MCF7	24	1180	55	0.257	0.319
SR-HSE	HA1E	6	814	52	0.234	0.248
SR-HSE	HepG2	6	547	30	0.269	0.252
SR-HSE	HT29	6	806	50	0.254	0.251
SR-HSE	MCF7	6	1077	64	0.248	0.245
SR-HSE	HA1E	24	697	43	0.231	0.238
SR-HSE	HT29	24	498	31	0.297	0.299
SR-HSE	MCF7	24	1065	62	0.239	0.254
SR-MMP	HA1E	6	723	129	0.588	0.646*
SR-MMP	HepG2	6	495	102	0.616	0.686*
SR-MMP	HT29	6	717	121	0.563	0.643*
SR-MMP	MCF7	6	958	143	0.527	0.635*
SR-MMP	HA1E	24	613	111	0.571	0.681*
SR-MMP	HT29	24	430	61	0.547	0.553
SR-MMP	MCF7	24	947	141	0.529	0.659*
SR-p53	HA1E	6	884	89	0.361	0.404
SR-p53	HepG2	6	603	64	0.344	0.395
SR-p53	HT29	6	877	85	0.338	0.427
SR-p53	MCF7	6	1158	100	0.295	0.397*
SR-p53	HA1E	24	757	76	0.350	0.381
SR-p53	HT29	24	533	57	0.374	0.399
SR-p53	MCF7	24	1142	98	0.294	0.508*